

Dornase alfa in mechanically ventilated children with bronchiolitis: A retrospective cohort study.

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The utility of the mucolytic dornase alfa in bronchiolitis has not been established, yet it is commonly used. The objective of this study was to compare outcomes of dornase alfa to standard of care treatments for bronchiolitis in mechanically ventilated pediatric patients. This was a retrospective, cohort study conducted at a single-center children's hospital evaluating pediatric patients with a diagnosis of bronchiolitis that required hospitalization and mechanical ventilation from January 1, 2010 to December 31, 2019. The primary outcome evaluated was length of time on mechanical ventilation. Secondary outcomes were pediatric intensive care unit (PICU) length of stay and length of hospitalization. Multiple linear regressions were used to assess the association of age, oxygen saturation index (OSI), positive end-expiratory pressure values, blood pH levels, respiratory syncytial virus status, and the use of other mucolytics, bronchodilator therapy, or chest physiotherapy treatment. Seventy-two patients were included in the study with 41 patients who were treated with dornase alfa. The patients who received dornase alfa had an average of 33.04 h longer on mechanical ventilation than those who did not ($p = 0.0487$). On average, they also had longer PICU and hospital stays by 2.05 days ($p = 0.053$) and 2.74 days ($p = 0.02$), respectively. In this study, pediatric patients who received dornase alfa had higher baseline OSI measurements than those who received standard of care, which impacted the primary outcome of time on mechanical ventilation and secondary outcome of time in the PICU. However, OSI, or any other variable, did not significantly affect results for the other secondary outcome of length of hospitalization. This study supports existing evidence that dornase alfa is not beneficial for bronchiolitis in pediatric patients, even in severe cases. Further prospective, randomized controlled trials are necessary to validate these outcomes.

Advances in the treatment and prevention of severe viral bronchiolitis.

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Viral bronchiolitis continues to be a major public health problem. Prevention of severe RSV-associated bronchiolitis has been achieved in high-risk infants by passive administration of the humanized monoclonal anti-RSV-F antibody, palivizumab. Development of more potent anti-RSV neutralizing antibodies is underway, and vaccine development continues to progress. Supportive therapy is the mainstay for the treatment of established infection while new antiviral and anti-inflammatory strategies are being explored in preclinical trials.

Safety and pharmacokinetics of palivizumab therapy in children hospitalized with respiratory syncytial virus infection.

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Respiratory syncytial virus (RSV) infection represents a major cause of pediatric respiratory hospitalizations. Limited treatment options exist. Palivizumab is a humanized monoclonal IgG1 antibody to the fusion protein of RSV that is highly active against RSV A and B strains. A phase I/II, multicenter, randomized, double-blind, placebo-controlled, escalating dose clinical trial to describe the safety,

tolerance, pharmacokinetics and clinical outcome of a single intravenous dose of palivizumab in previously healthy children hospitalized with acute RSV infection. Fifty-nine children ≤ 2 years of age received study drug. Sixteen children received 5 mg/kg of palivizumab ($n = 8$) or placebo ($n = 8$); 43 received 15 mg/kg of palivizumab ($n = 22$) or placebo ($n = 21$). Adverse events judged to be related to study drug were seen in one 5-mg/kg palivizumab patient and one 15-mg/kg palivizumab patient. These events were transient or consistent with progression of RSV disease. No discontinuations of study drug infusion because of adverse events occurred. Mean serum concentrations of palivizumab in the 5- and 15-mg/kg groups, respectively, were 61.2 and 303.4 microg/mL at 60 min and 11.2 and 38.4 microg/mL after 30 days. There were no significant differences in clinical outcomes between placebo and palivizumab groups for either dose. Intravenous palivizumab was safe and well-tolerated in children hospitalized with RSV disease. A single 15-mg/kg dose achieved serum palivizumab concentrations above the 25- to 30-microg/mL concentration associated with 2-log reduction of pulmonary RSV titer in the cotton rat model.

[Practical pediatric aspects of palivizumab].

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Palivizumab (Synagis) is a monoclonal antibody directed against the respiratory syncytial virus (RSV), for reducing mortality and morbidity in infants at risk of cardio-respiratory impairment due to bronchiolitis: 1. prematurity less than 28 weeks and less than 1 year of age; 2. between 28 and 32 weeks plus mechanical ventilation and less than 6 months of age; 3. chronic lung deficiency and less than 2 years of age; 4. congenital cardiopathy with either desaturation, pulmonary hypertension or cardiac failure. Another group of infants is those having a severe immunodeficiency. These infants are listed in a hospital recognized to have a competence in neonatal intensive care or a cardio-thoracic care program. The specialist in those disciplines prescribe the palivizumab which is delivered by the pharmacy of the competent hospital. The infant receives it by IM route at a dose of 15 mg/kg, monthly between October or November and February or March. Reduction of mortality and morbidity have been observed in the infants at risk. However, this costly pharmacological preventive approach needs to come after other simple preventive measures such as avoiding contact with potential carriers of nasal viruses and passive smoking, for bronchiolitis is not solely due to RSV.

[Respiratory syncytial virus vaccines].

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RESPIRATORY SYNCYTIAL VIRUS VACCINES. Respiratory syncytial virus (RSV) is responsible for lower respiratory infections, particularly in children under five years of age (acute infant bronchiolitis) and the elderly over 60. Monoclonal antibodies (palivizumab and nirsevimab) are used to prevent bronchiolitis. Four types of vaccine are currently under development: subunit vaccines composed of recombinant proteins or viral pseudoparticles, messenger RNA vaccines, recombinant vector vaccines and live attenuated vaccines. They are indicated for pregnant women to protect infants against bronchiolitis in the first months of life, and for people over 60 or with comorbidities.

Palivizumab for respiratory syncytial virus prophylaxis.

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Palivizumab, a humanized monoclonal antibody, has been approved by the FDA to prevent severe lower respiratory tract infections caused by RSV in high-risk patients. Prophylaxis of RSV infections with palivizumab requires monthly injections (15 mg/kg) during the RSV season. In the IMpact-RSV study, hospitalizations resulting from RSV decreased by 55% in the palivizumab treatment group. Palivizumab has also been shown to decrease the number of days with moderate or severe RSV infection, with an increased oxygen requirement, and ICU admissions. Palivizumab has been shown to be well tolerated with minimal adverse effects. The most frequently reported adverse effects were fever and minor injection site reactions. Determination of which patients should receive RSV prophylaxis should take into consideration all risk factors. Recommendations for RSV prophylaxis with RSV-IGIV and palivizumab have been published by the American Academy of Pediatrics. To date, no studies directly comparing RSV-IGIV and palivizumab have been conducted. Neither product is recommended in children with congenital heart disease.

Nirsevimab for the prevention of respiratory syncytial virus disease in children. Statement of the Spanish Society of Paediatric Infectious Disease (SEIP).

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Nirsevimab, a monoclonal antibody for the prevention of disease caused by respiratory syncytial virus (RSV), has recently been approved for use in Europe and Spain. To provide recommendations for the administration of nirsevimab for prevention of RSV disease. The approach chosen to develop these recommendations involved a critical review of the literature and the use of the Delphi and GRADE methods. An expert group was formed. The group engaged in three rounds to define the questions, express support or opposition, grade recommendations and establish the agreement or disagreement with the conclusions. In the general neonatal population, routine administration of nirsevimab is recommended to reduce the frequency of illness and hospitalisation for bronchiolitis and RSV lower respiratory tract infection. Nirsevimab is recommended for all infants born in high-incidence RSV season and infants aged less than 6 months at the season onset. In infants born preterm between 29 and 35 weeks of gestation, with haemodynamically significant heart disease or with chronic lung disease, routine administration of nirsevimab is recommended to reduce the incidence of disease and hospitalisation due to bronchiolitis and RSV lower respiratory tract infection. In patients in whom palivizumab is currently indicated, its substitution by nirsevimab is recommended to reduce the burden of bronchiolitis. Routine administration of nirsevimab to all infants aged less than 6 months born during the RSV season or aged less than 6 months at the start of the winter season is recommended to reduce the burden of disease and the frequency of hospitalization due to bronchiolitis.

Eligibility for palivizumab prophylaxis in a cohort of children with severe bronchiolitis.

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In 2014, the American Academy of Pediatrics (AAP) updated their recommendations for palivizumab prophylaxis for children who are at high risk for severe respiratory syncytial virus (RSV) infection. To

investigate the potential impact of the more restrictive 2014 criteria on the eligibility for palivizumab prophylaxis, we applied the 2012 and 2014 AAP recommendations for palivizumab prophylaxis to a multicenter cohort of 2207 US children hospitalized for bronchiolitis. According to the 2012 AAP recommendations, 215 children (9.7%) were eligible for palivizumab prophylaxis, while 140 children (6.3%) would have been eligible based on the 2014 updated recommendations (34.9% relative decrease; 95%CI: 28.5-41.7%). The decrease was largely driven by the restriction of eligibility to preterm infants with gestational age <29 weeks. Further development of and refinement of cost-effective approaches for the prevention of severe RSV infection are needed.

Revised indications for the use of palivizumab and respiratory syncytial virus immune globulin intravenous for the prevention of respiratory syncytial virus infections.

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Palivizumab and Respiratory Syncytial Virus Immune Globulin Intravenous (RSV-IGIV) are licensed by the Food and Drug Administration for use in preventing severe lower respiratory tract infections caused by respiratory syncytial virus (RSV) in high-risk infants, children younger than 24 months with chronic lung disease (formerly called bronchopulmonary dysplasia), and certain preterm infants. This statement provides revised recommendations for administering RSV prophylaxis to infants and children with congenital heart disease, for identifying infants with a history of preterm birth and chronic lung disease who are most likely to benefit from immunoprophylaxis, and for reducing the risk of RSV exposure and infection in high-risk children. On the basis of results of a recently completed clinical trial, prophylaxis with palivizumab is appropriate for infants and young children with hemodynamically significant congenital heart disease. RSV-IGIV should not be used in children with hemodynamically significant heart disease. Palivizumab is preferred for most high-risk infants and children because of ease of intramuscular administration. Monthly administration of palivizumab during the RSV season results in a 45% to 55% decrease in the rate of hospitalization attributable to RSV. Because of the large number of infants born after 32 to 35 weeks' gestation and because of the high cost, immunoprophylaxis should be considered for this category of preterm infants only if 2 or more risk factors are present. High-risk infants should not attend child care during the RSV season when feasible, and exposure to tobacco smoke should be eliminated.

Analysis of respiratory syncytial virus preclinical and clinical variants resistant to neutralization by monoclonal antibodies palivizumab and/or motavizumab.

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Palivizumab is a US Food and Drug Administration-approved monoclonal antibody for the prevention of respiratory syncytial virus (RSV) lower respiratory disease in high-risk infants. Motavizumab, derived from palivizumab with enhanced antiviral activity, has recently been tested in humans. Although palivizumab escape mutants have been generated in the laboratory, the development of resistant RSV in patients receiving palivizumab has not been reported previously. We generated palivizumab and motavizumab escape mutants in vitro and examined the development of resistant mutants in RSV-breakthrough patients receiving immunoprophylaxis. The effect of these mutations on neutralization by palivizumab and motavizumab and in vitro fitness was studied. Antibody-resistant RSV variants selected in vitro had

mutations at position 272 of the fusion protein, from lysine to asparagine, methionine, threonine, glutamine, or glutamate. Variants containing mutations at positions 272 and 275 were detected in breakthrough patients. All these variants were resistant to palivizumab, but only the glutamate variant at position 272 demonstrated resistance to motavizumab. Mixtures of wild-type and variant RSV soon lost the resistant phenotype in the absence of selection. Resistant RSV variants were detected in a small subset (~ 5%) of RSV breakthrough cases. The fitness of these variants was impaired, compared to wild-type RSV.
